

Evaluation of Classifiers: Step-by-Step Solution

Confusion Matrix Analysis & Precision vs. Recall Trade-offs in Medical Diagnostics

PROBLEM SUMMARY & GIVEN DATA:

- Context: Medical diagnostic classifier for cancer detection (Positive Class = 'yes', Negative Class = 'no').
- Total Sample Size: N = 10,000 patients.
- Actual Cancer Patients (yes): 300 (3% prevalence — heavily imbalanced).
- Actual Non-Cancer Patients (no): 9,700 (97% prevalence).

Step 1: Mapping the Confusion Matrix

To analyze any classification model, we first map each quadrant of the given confusion matrix to its standard statistical terminology: True Positives (TP), False Negatives (FN), False Positives (FP), and True Negatives (TN).

Actual Class	Predicted: yes	Predicted: no	Actual Total	Class Recognition Rate	Terminology Mapping
Actual = yes	90 (TP)	210 (FN)	300	$90 / 300 = 30.00\%$	TP = 90, FN = 210
Actual = no	140 (FP)	9560 (TN)	9700	$9560 / 9700 = 98.56\%$	FP = 140, TN = 9560
Total Predicted	230 (TP + FP)	9770 (FN + TN)	10,000 (Total N)	$9650 / 10000 = 96.50\%$	Overall Accuracy

Fundamental Definitions for Students:

- True Positive (TP = 90): Patient HAS cancer, and model correctly predicted 'yes'.
- False Negative (FN = 210): Patient HAS cancer, but model incorrectly predicted 'no' (Missed cancer diagnosis).
- False Positive (FP = 140): Patient is HEALTHY, but model incorrectly predicted 'yes' (False alarm).
- True Negative (TN = 9560): Patient is HEALTHY, and model correctly predicted 'no'.

Question 1: Calculate Precision, Recall, F-Score, Error-rate

1. Precision (Positive Predictive Value)

Precision answers: 'Out of all patients whom the classifier flagged as having cancer, how many actually have cancer?'

Formula: Precision = $TP / (TP + FP)$

Calculation: Precision = $90 / (90 + 140) = 90 / 230 \approx 0.391304$

Result: Precision = 39.13%

2. Recall (Sensitivity / True Positive Rate / Hit Rate)

Recall answers: 'Out of all actual cancer patients in the population, how many did the classifier successfully catch?'

Formula: Recall = $TP / (TP + FN)$

Calculation: Recall = $90 / (90 + 210) = 90 / 300 = 0.3000$

Result: Recall = 30.00% (Note: Also given in table as Class 'yes' Recognition = 30.00%)

3. F-Score (F1-Score / Harmonic Mean)

The F-Score balances Precision and Recall into a single metric using their harmonic mean. It is especially vital when dealing with imbalanced datasets where accuracy is misleading.

Formula: F-Score = $2 \times (Precision \times Recall) / (Precision + Recall)$

Calculation: F-Score = $2 \times (0.391304 \times 0.3000) / (0.391304 + 0.3000)$

$= 2 \times (0.117391) / (0.691304) = 0.234782 / 0.691304 \approx 0.33962$

Result: F-Score = 0.3396 (or 33.96%)

4. Error-Rate (Misclassification Rate)

Error-rate measures the overall proportion of incorrect predictions across the entire dataset.

Formula: Error-Rate = $(FP + FN) / Total = 1 - Accuracy$

Calculation: Error-Rate = $(140 + 210) / 10,000 = 350 / 10,000 = 0.0350$

Alternatively: Accuracy = $(TP + TN) / Total = (90 + 9560) / 10,000 = 96.50\%$

$$\text{Error-Rate} = 1 - 0.9650 = 0.0350$$

Result: Error-Rate = 3.50% (Overall Accuracy = 96.50%)

Performance Metric	Formula	Computed Value
Precision	$TP / (TP + FP)$	39.13% (90 / 230)
Recall (Sensitivity)	$TP / (TP + FN)$	30.00% (90 / 300)
F1-Score	$2 \times (\text{Prec} \times \text{Rec}) / (\text{Prec} + \text{Rec})$	0.3396 (33.96%)
Overall Accuracy	$(TP + TN) / \text{Total}$	96.50% (9650 / 10000)
Error-Rate	$(FP + FN) / \text{Total} = 1 - \text{Accuracy}$	3.50% (350 / 10000)

Question 2: Scenarios Where PRECISION Is Preferred Over Recall

Precision is prioritized when the cost, risk, or harm of a False Positive (predicting 'yes' when actual is 'no') is significantly higher than the cost of a False Negative.

Medical & Practical Scenarios Preferring High Precision:

1. High-Risk / Invasive Follow-up Interventions:

- If a 'yes' prediction leads immediately to major invasive surgery, intensive chemotherapy, or organ removal with severe irreversible side effects.
- In this case, diagnosing a healthy patient as positive (False Positive) subjects them to dangerous, unnecessary trauma. We need near-100% precision before operating.

2. Extremely Expensive or Scarce Confirmatory Diagnostics:

- If positive screening requires an extremely expensive specialized test (e.g., \$10,000 PET scan or rare specialist biopsy) with severely limited hospital capacity.
- High False Positives would overwhelm hospital resources and deny care to genuinely sick individuals.

3. Severe Psychological Trauma & Stigma:

- When communicating severe terminal diagnoses where false alarms cause immense mental anguish, depression, or irreversible life decisions.

4. General Non-Medical Analogies:

- Spam Filtering: Sending an important job offer email to Spam (False Positive) is worse than letting one

spam email into the inbox.

- Criminal Justice: 'Better that ten guilty persons escape than that one innocent suffer' (minimizing False Positives / wrongful convictions).

Question 3: Scenarios Where RECALL Is Preferred Over Precision

Recall is prioritized when the cost of a False Negative (missing a positive case) is catastrophic or fatal, while the cost of a False Positive is minimal and easily resolved.

Medical & Practical Scenarios Preferring High Recall:

1. Primary Routine Cancer Screening (Mammography, Pap Smears, Low-Dose CT):

- If a cancer patient is told they are healthy (False Negative, 210 patients in this example), their cancer will progress undetected to late stages, likely resulting in avoidable death.
- Conversely, a False Positive only results in a harmless, inexpensive secondary follow-up test (e.g., ultrasound or blood test) to verify health.
- Missing cancer is fatal; a false alarm is merely an inconvenience.

2. Highly Contagious Disease Outbreak Screening (e.g., COVID-19, Ebola, Tuberculosis):

- Missing an infected individual (False Negative) allows super-spreader events to infect entire communities.
- Quarantining a healthy individual for 24 hours while conducting a confirmatory PCR test (False Positive) is an acceptable low-cost precaution.

3. Airport Security & Threat Detection:

- Missing a weapon or explosive on an aircraft (False Negative) is fatal.
- Manually opening and inspecting a harmless bag (False Positive) takes 30 seconds and has negligible cost.

4. Critical Machine / Engine Failure Monitoring:

- Missing a turbine fracture before flight causes a crash; performing an extra inspection routine is routine and safe.

- High Accuracy Paradox: The model exhibits 96.50% overall accuracy, yet it misses 70% of cancer cases (Recall = 30.00%) and over 60% of its positive alerts are false alarms (Precision = 39.13%).

- In clinical deployment as a cancer screening tool, this model is dangerously inadequate because a 30% Recall means 210 out of 300 cancer patients go completely untreated.

- Rule of Thumb: In screening life-threatening diseases, always maximize RECALL first (aim for > 95%), even at the expense of lower precision, and use multi-tier confirmatory testing.

